

HW 4

March 30, 2023

Problem 0.1 (Chapter 10, problem 1). For each of the following pairs (a, b) of integers, find the highest common factor $d = \text{hcf}(a, b)$, and find integers s, t such that $d = sa + tb$.

(i) $a = 17, b = 29$

(ii) $a = 552, b = 713$

(iii) $a = 345, b = 299$

Proof. Part (i)

$$\text{hcf}(17, 29) = 1$$

$$1 = 12 \cdot 17 - 29 \cdot 7$$

Part (ii)

$$\text{hcf}(552, 713) = 23$$

$$23 = 22 \cdot 552 - 17 \cdot 713$$

Part (iii)

$$\text{hcf}(345, 299) = 23$$

$$-6 \cdot 345 + 7 \cdot 299 = 23$$

□

Problem 0.2 (Chapter 10, problem 4). (a) Show that for all positive integer n , $\text{hcf}(6n+8, 4n+5) = 1$.

(b) Suppose a, b are integers such that $a|b$ and $b|a$. Prove that $a = \pm b$.

(c) Suppose s, t, a, b are integers such that $sa + tb = 1$. Show that $\text{hcf}(a, b) = 1$.

Proof. (a) (1 point) Lets start doing the Euclidean algorithm. We get

$$6n + 8 = 1 \cdot (4n + 5) + (2n + 3)$$

$$4n + 5 = 1 \cdot (2n + 3) + (2n + 2)$$

$$2n + 3 = 1 \cdot (2n + 2) + 1$$

$$2n + 2 = (2n + 2) \cdot 1 + 0$$

Therefore the hcf is 1.

(b) If $a|b$, then $b = ak$ for some k . If $b|a$, then $a = bl$ for some l . Combining these, we get $a = bl = (ak)l = a(kl)$. We know a is nonzero, so we can divide to conclude that $1 = kl$. The only way this is possible is if k and l are each ± 1 . Therefore $a = \pm b$.

(c) Suppose that $d = \text{hcf}(a, b)$. Then d divides a and b , so d divides $sa + tb$ for any integers s, t . But by assumption we have $sa + tb = 1$, so $d|1$. Therefore $d = 1$. □

Problem 0.3 (Chapter 10, problem 9). Let a, b be coprime positive integers. Prove that for any integer n there exist integers s, t with $s > 0$ such that $sa + tb = n$.

Proof. We know from lecture that there exist integers s, t such that $sa + tb = 1$.

Furthermore, we claim that we can find such s, t so that $s > 0$. To see this, suppose that $s < 0$. Note that we have

$$bm \cdot a - am \cdot b = 0$$

for any integer m . Take m large enough so that $bm + s > 0$. Now add $bm \cdot a - am \cdot b$ to $sa + tb = 1$ to get

$$(bm + s) \cdot a + (t - am) \cdot b = 1$$

We have $bm + s > 0$, as we wanted.

Now, say we have found s, t such that $sa + tb = 1$ and $s > 0$. Just multiply both sides by n to get

$$(sn)a + (tn)b = n$$

We have $sn > 0$, so we are done. □